Exercise 2.7 (One Bounded Product Implies Another)

Show that for nonnegative x, y, and z one has the implication

$$1 \leq xy\cancel{x}_{\,yz} \implies 8 \leq (1+x)(1+y)(1+z). \qquad (2.28)$$

Can you also propose a generalization?

$x, y, z \geqslant 0 \; : \; xyz \geqslant 1 \Rightarrow (1+x)(1+y)(1+z) \geqslant 8$

Demonstração.

$MA \geqslant MG : \; \dfrac{x+y+z}{3} \geqslant \sqrt[3]{xyz}$

$$\boxed{x, y, z \geqslant 0 \text{ e } \underbrace{xyz \geqslant 1} \Rightarrow x, y, z > 0}$$

$\dfrac{1+x}{2} \geqslant \sqrt{1 \cdot x} = \sqrt{x}$ - - - - (1) $\quad [MA \geqslant MG, n = 2]$

$\dfrac{1+y}{2} \geqslant \sqrt{1 \cdot y} = \sqrt{y}$. — . (2)

$\dfrac{1+z}{2} \geqslant \sqrt{1 \cdot z} = \sqrt{z}$ - - - - (3)

$$\left(\frac{1+x}{2}\right)\left(\frac{1+y}{2}\right)\left(\frac{1+z}{2}\right) \geqslant \sqrt{x} \cdot \sqrt{y} \cdot \sqrt{z} = \sqrt{xyz}$$

$$(1+x)(1+y)(1+z) \geqslant 8\sqrt{xyz} \geqslant 8 \cdot 1 = 8$$

$xyz \geqslant 1 \Rightarrow (1+x)(1+y)(1+z) \geqslant 8$ ∎

$$\prod_{k=1}^{n} x_k \geqslant 1 \Rightarrow \prod_{k=1}^{n} (1+x_k) \geqslant 2^n$$

Demonstração.

$\dfrac{1+x_k}{2} \overset{MA \geqslant MG}{\geqslant} \sqrt{x_k}, \quad k = 1, 2, \ldots, n$

x

$$\prod_{k=1}^{n} \underbrace{\frac{(1+x_k)}{2^n}} = \frac{1}{2^n} \prod_{k=1}^{n} (1+x_k) \geqslant \sqrt{x_1} \cdot \sqrt{x_2} \cdots \sqrt{x_n}$$

$$\prod_{k=1}^{n} (1+x_k) \geqslant 2^n \sqrt{x_1 \cdot x_2 \cdots x_n}$$

$$\prod_{k=1}^{n} x_k \geqslant 1 \Rightarrow \prod_{k=1}^{n} (1+x_k) \geqslant 2^n \quad \blacksquare$$

$$\prod_{k=1}^{n} x_k \geqslant a^n \Rightarrow \prod_{k=1}^{n} (1+x_k) \geqslant \boxed{\ ?\ }$$

↳ ínfimo

P_1 : $\min\{ x_1 \cdot x_2 \cdots x_n \ : \ a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \}$

P_2 : $\min\{ a_1x_1 + a_2x_2 + \cdots + a_nx_n \ : \ x_1 \cdot x_2 \cdots x_n = c \}$

$$a_1x_1 = a_2x_2 = \cdots = a_nx_n$$

→ Verdadeiro

$$\prod_{k=1}^{n} (1+x_k) \geqslant \underbrace{(1+a)^n \geqslant 2^n \cdot a^{n/2}}$$

$$\prod_{k=1}^{n} (1+x_k) \geqslant 2^n \cdot a^{n/2}$$

$$\left.\begin{aligned} lhs &= (1+1)^n = 2^n \\ rhs &= 2^n \cdot 1^{n/2} = 2^n \end{aligned}\right\} \to \text{para } a=1 \Rightarrow (1+a)^n = 2^n$$

Desigualdade de Jensen

$$t_k = \ln x_k \Leftrightarrow x_k = e^{t_k}$$

f convexa

$$\boxed{f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i)} \to \text{Jensen}$$

$$\boxed{f\left(\sum_{i=1}^{n}\lambda_i x_i\right) \leq \sum_{i=1}^{n}\lambda_i f(x_i)} \to \text{Jensen}$$

$$\boxed{\sum_{k=1}^{n} \ln(1+x_k) \geq n\ln(1+a) \Rightarrow \prod_{k=1}^{n}(1+x_k) \geq (1+a)^n}$$

$x_k = a$:. $x_1 = x_2 = \dots = x_n = a$

$$\prod_{k=1}^{n}(1+x_k) \xrightarrow[k=1,\dots,n]{x_k=a} \prod_{k=1}^{n}(1+x_k) = \prod_{k=1}^{n}(1+a) = (1+a)(1+a)\dots(1+a) = (1+a)^n$$

Se $x_k = a, \forall k = 1,\dots,n$ então $\prod_{k=1}^{n}(1+x_k) = (1+a)^n$

$n=2$:. $(1+x_1)(1+x_2) = 1 + (x_1+x_2) + x_1 \cdot x_2 \geq 1 + 2\sqrt{x_1 \cdot x_2} + x_1 \cdot x_2$

MA $\geq$ MG: $x_1 + x_2 \geq 2\sqrt{x_1 \cdot x_2}$

$$(1+x_1)(1+x_2) \geq 1 + 2\sqrt{x_1x_2} + x_1x_2 = (1+\sqrt{x_1x_2})^2$$

$$(1+x_1)(1+x_2) \geq (1+\sqrt{x_1x_2})^2$$

$n=3$:. $(1+x_1)(1+x_2)(1+x_3) = (1+x_2+x_1+x_1x_2)(1+x_3)$

$= (1+x_3+x_2+x_2x_3+x_1+x_1x_3+x_1x_2+x_1x_2x_3)$

$$\prod_{k=1}^{3}(1+x_k) = 1 + (x_1+x_2+x_3) + (x_1 \cdot x_2 + x_1x_3 + x_2 \cdot x_3) + x_1 \cdot x_2 \cdot x_3$$

MA $\geq$ MG:

$x_1 + x_2 + x_3 \geq 3\sqrt[3]{x_1x_2x_3}$

$$x_1x_2 + x_1x_3 + x_2x_3 \geq 3\sqrt[3]{x_1 \cdot x_2 \cdot x_1x_3 \cdot x_2x_3} = 3(x_1x_2x_3)^{2/3}$$

$$\prod_{k=1}^{3}(1+x_k) \geq 1 + 3\sqrt[3]{x_1x_2x_3} + 3(x_1x_2x_3)^{2/3} + x_1x_2x_3$$

$y^3 = x_1 x_2 x_3 \Rightarrow y^2 = (x_1x_2x_3)^{2/3}$, $y = \sqrt[3]{x_1x_2x_3}$

$k=1$

$$y^3 = x_1 x_2 x_3 \Rightarrow y^2 = (x_1 x_2 x_3)^{2/3}, \quad y = \sqrt[3]{x_1 x_2 x_3}$$

$$\prod_{k=1}^{3} (1+x_k) \geq 1 + 3y + 3y^2 + y^3 = (1+y)^3$$

$$\prod_{k=1}^{3} (1+x_k) \geq (1+y)^3, \quad y = \sqrt[3]{x_1 x_2 x_3}$$

$$\vdots$$

$$\prod_{k=1}^{n} (1+x_k) \geq \left[1 + \left(\prod_{k=1}^{n} x_k\right)^{1/n}\right]^n \geq (1+a)^n$$

$$\prod_{k=1}^{n} x_k \geq a^n \Rightarrow \left(\prod_{k=1}^{n} x_k\right)^{1/n} \geq a \qquad y^n = \prod_{k=1}^{n} x_k$$